

NEHA VALUTHUNDIL SHAJI

Sharjah, UAE | nehavshaji@gmail.com | <https://www.linkedin.com/in/nehavshaji> | <https://github.com/Nehashaji> | <https://nehashaji-portfolio.vercel.app>

PROFESSIONAL SUMMARY

BSc (Hons) Creative Computing graduate with practical experience in machine learning, computer vision, data analytics, and business intelligence through academic and independent projects. Skilled in Python, SQL, Scikit-learn, TensorFlow, and Power BI, with experience across data preprocessing, exploratory analysis, predictive modelling, model evaluation, and interactive data visualization. Seeking to develop advanced technical and research expertise through postgraduate study in **MSc Data Science and Artificial Intelligence**.

EDUCATION

- **BSc (Hons) Creative Computing - First Class Honours**

Bath Spa University, RAK Campus | 2023-2026

Relevant coursework: Machine Learning, Emerging Technologies, Research Project

TECHNICAL SKILLS

Programming & Data: Python (Pandas, NumPy), SQL (MySQL)

Machine Learning & AI: Supervised Learning, Classification, Regression, Deep Learning, CNNs, Transfer Learning (MobileNetV2), Model Evaluation, Hyperparameter Tuning

Frameworks & Platforms: Scikit-learn, TensorFlow

Data Visualisation & BI: Power BI (DAX, Power Query), Matplotlib, Seaborn

Deployment & Tools: Gradio, Git/GitHub, Hugging Face Spaces, Jupyter/Kaggle Notebooks, Excel

Additional: React, Node.js, MongoDB, Figma

DATA SCIENCE & AI PROJECTS

Weather Condition Classification System | *Computer Vision - Python, TensorFlow, CNN, MobileNetV2*

- Built an end-to-end image classification pipeline to categorise outdoor images into four weather conditions (Cloudy, Rainy, Shine, Sunrise).
- Benchmarked a baseline CNN against a MobileNetV2 transfer-learning model, applying data augmentation and fine-tuning to improve generalisation.
- Achieved ~87% validation accuracy; evaluated performance using precision, recall, F1-score and confusion matrix analysis.
- Deployed the trained model as a live web application on Hugging Face Spaces, providing real-time predictions with confidence scores.
- **Kaggle Link:** <https://www.kaggle.com/code/neha005/weather-image-classification-ml-project>
- **Live Demo Link:** <https://huggingface.co/spaces/NehaShaji/weather-classifier>

Olist E-Commerce Analytics Dashboard | *Business Intelligence - SQL, Power BI, DAX, Power Query*

- Analysed 99,441 real-world marketplace orders from the Brazilian Olist e-commerce dataset, validating relational data integrity prior to analysis.
- Wrote advanced SQL (CTEs, window functions - RANK, ROW_NUMBER, LAG, NTILE and cumulative aggregations) to build customer segmentation and cohort retention analyses.
- Designed a four-page interactive Power BI dashboard (Executive Overview, Customer Analysis, Products & Sellers, Delivery Performance) using custom DAX measures.
- Surfaced key business insights, including that 96.88% of customers were one-time buyers and that the top 10 sellers (0.3% of sellers) drove 13.15% of item revenue.
- **Github Link:** <https://github.com/Nehashaji/Olist-ecommerce-analytics>

Online Purchase Prediction | *Classification - Python, Scikit-learn*

- Preprocessed and analysed online shopper behaviour data, handling class imbalance through oversampling and encoding categorical/boolean features.

- Trained and compared Logistic Regression, KNN, Decision Tree, and Random Forest classifiers, tuning hyperparameters to balance generalisation and overfitting.
- Selected Random Forest as the final model for its consistent precision/recall balance and generalisation ability, despite a tuned KNN model achieving a marginally higher cross-validated F1-score.
- **Kaggle Link:** <https://www.kaggle.com/code/neha005/02-classification-task>

Sports Car Price Prediction | *Regression - Python, Scikit-learn*

- Cleaned and preprocessed a real-world automotive dataset, resolving skewed distributions and missing values via median imputation.
- Built and compared Linear Regression and Decision Tree Regression models, using cross-validation and residual analysis to assess reliability.
- **Kaggle Link:** <https://www.kaggle.com/code/neha005/01-regression-task>

CERTIFICATIONS

- **Kaggle** - [Intermediate Machine Learning](#)
- **Kaggle** - [Intro to Machine Learning](#)
- **LinkedIn Learning** - [Machine Learning Foundations: Linear Algebra](#)
- **LinkedIn Learning** - [Probability Foundations for Data Science](#)
- **IBM** - [Python for Data Science](#)
- **Kaggle** - [Intro to SQL](#)
- **Cisco Networking Academy** - [Data Analytics Essentials](#)
- **LinkedIn Learning** - [Complete Guide to Power BI for Data Analysts](#)
- **Microsoft & LinkedIn Learning** - [Career Essentials in Generative AI](#)